ICD-10, ICD-9, OPCS-4, and self-report codes used for disease definitions.

Outcome	ICD-10	ICD-9	OPCS-4	Non-cancer illness code[1]	Operation code[1]	Exclusion criteria[2]
Coronary artery disease (broad)	I21: I21.0, I21.1, I21.2, I21.3, I21.4, I21.9; I22: I22.0, I22.1, I22.8, I22.9; I23: I23.0, I23.1, I23.2, I23.3, I23.5, I23.6, I23.8; I24: I24.0, I24.1, I24.8, I24.9; I25: I25.0, I25.1, I25.2, I25.3, I25.4, I25.5, I25.6, I25.8, I25.9;	410: 410.9; 411: 411.9; 412: 412.9; 413: 413.9; 414: 414.0, 414.8, 414.9;	K40: K40.1, K40.2, K40.3, K40.4, K40.8, K40.9; K41: K41.1, K41.2, K41.3, K41.4, K41.8, K41.9; K45: K45.1, K45.2, K45.3, K45.4, K45.5, K45.6, K45.8, K45.9; K49: K49.1, K49.2, K49.3, K49.4, K49.8, K49.9; K50: K50.2; K75: K75.1, K75.2, K75.3, K75.4, K75.8, K75.9;	1075 ("heart attack/myocardial infarction");	1070 ("coronary angioplasty (ptca) +/- stent"), 1095 ("coronary artery bypass grafts (cabg)"), 1523 ("triple heart bypass");	None
Myocardial infarction	I21: I21.0, I21.1, I21.2, I21.3, I21.4, I21.9; I22: I22.0, I22.1, I22.8, I22.9; I23: I23.0, I23.1, I23.2, I23.3, I23.5, I23.6, I23.8; I25: I25.2;	410: 410.9; 411: 411.9; 412: 412.9;	NA	1075 ("heart attack/myocardial infarction");	NA	None
Abdominal aortic aneurysm	I71: I71.3, I71.4, I71.5, I71.6;	441: 441.4;	L18: L18.2, L18.3, L18.4, L18.5, L18.6, L18.8, L18.9; L19: L19.1, L19.2, L19.3, L19.4, L19.5, L19.6, L19.8, L19.9; L25: L25.4; L27: L27.1, L27.2, L27.3, L27.4, L27.5, L27.6, L27.8, L27.9; L28: L28.1, L28.2, L28.3, L28.4, L28.5, L28.6, L28.8, L28.9; L46: L46.4;	1492 ("aortic aneurysm"), 1591 ("aortic aneurysm rupture"), 1592 ("aortic dissection");	1104 ("aortic aneurysm/repair or stent");	None
All-cause mortality[3]	Any cause (death registry, date of death)	NA	NA	NA	NA	NA
Alzheimer's disease	F00: F00.0, F00.1, F00.2, F00.9; G30: G30.0, G30.1, G30.8, G30.9;	NA	NA	NA	NA	None
Anemia	D50: D50.0, D50.1, D50.8, D50.9; D51: D51.0, D51.1, D51.2, D51.3, D51.8, D51.9; D52: D52.0, D52.1, D52.8, D52.9; D53: D53.0, D53.1, D53.8, D53.9; D55: D55.0, D55.2, D55.9; D56: D56.0, D56.1, D56.2, D56.3, D56.4, D56.8, D56.9; D57: D57.0, D57.1, D57.2, D57.3, D57.8; D58: D58.0, D58.1, D58.2, D58.8, D58.9; D59: D59.0, D59.1, D59.2, D59.3, D59.4, D59.5, D59.6, D59.8, D59.9; D60: D60.0, D60.9; D61: D61.0, D61.1, D61.2, D61.3, D61.8, D61.9; D62; D63: D63.0, D63.8; D64: D64.0, D64.1, D64.2, D64.3, D64.4, D64.8, D64.9;	280: 280.90, 280.99; 281: 281.0, 281.2, 281.9; 282: 282.0, 282.7, 282.9; 283: 283.0; 284: 284.0, 284.89, 284.9; 285: 285.1, 285.81, 285.9;	NA	1446 ("anaemia"), 1330 ("iron deficiency anaemia"), 1331 ("pernicious anaemia"), 1332 ("aplastic anaemia");	NA	None
Anxiety disorders	F40: F40.0, F40.1, F40.2, F40.8, F40.9; F41: F41.0, F41.1, F41.2, F41.3, F41.8, F41.9; F42: F42.0, F42.1, F42.2, F42.8, F42.9; F43: F43.0, F43.1, F43.2, F43.8, F43.9; F44: F44.0, F44.1, F44.2, F44.3, F44.4, F44.5, F44.6, F44.7, F44.8, F44.9; F45: F45.0, F45.1, F45.2, F45.3, F45.4, F45.8, F45.9; F48: F48.0, F48.1, F48.8, F48.9;	300: 300.0, 300.1, 300.2, 300.3, 300.4, 300.5, 300.7, 300.9;	NA	1287 ("anxiety/panic attacks"), 1288 ("nervous breakdown"), 1469 ("post-traumatic stress disorder"), 1614 ("stress"), 1615 ("obsessive compulsive disorder (ocd)");	NA	None
Aortic stenosis	I35: I35.0, I35.2;	424: 424.1;	NA	1490 ("aortic stenosis");	NA	None
Asthma	J45: J45.0, J45.1, J45.8, J45.9; J46;	493: 493.0, 493.1, 493.9;	NA	1111 ("asthma");	NA	None
Atrial fibrillation	I48: I48.0, I48.1, I48.2, I48.3, I48.4, I48.9;	427: 427.3;	K57: K57.1; K62: K62.1, K62.2, K62.3, K62.4;	1471 ("atrial fibrillation"), 1483 ("atrial flutter");	1524 ("cardioversion");	None
Biliary and pancreatic disease	K80: K80.0, K80.1, K80.2, K80.4, K80.5, K80.8; K81: K81.0, K81.1, K81.8, K81.9; K82: K82.0, K82.1, K82.2, K82.3, K82.4, K82.8, K82.9; K83: K83.0, K83.1, K83.2, K83.3, K83.4, K83.5, K83.8, K83.9; K86: K86.0, K86.1, K86.2, K86.3, K86.8, K86.9;	574: 574.0, 574.1, 574.2, 574.3, 574.4, 574.5; 575: 575.0, 575.1, 575.3, 575.6, 575.8, 575.9; 576: 576.0, 576.1, 576.2, 576.4, 576.8, 576.9;	J18: J18.1, J18.2, J18.3, J18.4, J18.5, J18.8, J18.9; J38: J38.1, J38.2, J38.8, J38.9;	1162 ("cholelithiasis/gall stones"), 1163 ("cholecystitis"), 1160 ("bile duct obstruction/ascending cholangitis"), 1165 ("pancreatitis");	1455 ("cholecystectomy/gall bladder removal"), 1456 ("bile duct surgery"), 1528 ("gallstones removed");	None
Cardiovascular death[4]	I00–I99 (any ICD-10 chapter IX "I" code) as primary (underlying) cause of death	NA	NA	NA	NA	NA
Chronic kidney disease	I12: I12.0; I13: I13.1, I13.2; N18: N18.0, N18.1, N18.2, N18.3, N18.4, N18.5, N18.8, N18.9;	585: 585.9;	L74: L74.6; M01: M01.1, M01.2, M01.3, M01.4, M01.5, M01.8, M01.9; M02: M02.6, M02.7; M08: M08.4; M17: M17.2,	NA	NA	None

Outcome	ICD-10	ICD-9	OPCS-4	Non-cancer illness code[1]	Operation code[1]	Exclusion criteria[2]
			M17.4, M17.8, M17.9; X40: X40.1, X40.2, X40.3, X40.5, X40.6; X41: X41.1, X41.2; X42: X42.1;			
Chronic obstructive pulmonary disease	J43: J43.0, J43.1, J43.2, J43.8, J43.9; J44: J44.0, J44.1, J44.8, J44.9;	492: 492.9; 496: 496.9;	NA	1112 (“chronic obstructive airways disease/copd”), 1113 (“emphysema/chronic bronchitis”), 1472 (“emphysema”);	NA	None
Depression	F32: F32.0, F32.1, F32.2, F32.3, F32.8, F32.9; F33: F33.0, F33.1, F33.2, F33.3, F33.4, F33.8, F33.9; F41: F41.2;	300: 300.4; 311: 311.9;	NA	1286 (“depression”);	NA	ICD-10 – F20: F20.0, F20.1, F20.2, F20.3, F20.4, F20.5, F20.6, F20.8, F20.9; F21; F22: F22.0, F22.8, F22.9; F23: F23.0, F23.1, F23.2, F23.3, F23.8, F23.9; F24; F25: F25.0, F25.1, F25.2, F25.8, F25.9; F28; F29; F30: F30.0, F30.1, F30.2, F30.8, F30.9; F31: F31.0, F31.1, F31.2, F31.3, F31.4, F31.5, F31.6, F31.7, F31.8, F31.9; ICD-9 – 295: 295.0, 295.1, 295.3, 295.4, 295.5, 295.6, 295.7, 295.9; 296: 296.0, 296.1, 296.2, 296.3, 296.4, 296.5, 296.6, 296.9; Non-cancer illness – 1289 (“schizophrenia”), 1291 (“mania/bipolar disorder/manic depression”);
Epilepsy	G40: G40.0, G40.1, G40.2, G40.3, G40.4, G40.6, G40.7, G40.8, G40.9;	345: 345.09, 345.10, 345.19, 345.2, 345.3, 345.4, 345.50, 345.59, 345.7, 345.9;	NA	1264 (“epilepsy”);	NA	ICD-10 – G40: G40.5;
Glaucoma	H40: H40.0, H40.1, H40.2, H40.3, H40.4, H40.5, H40.6, H40.8, H40.9;	365: 365.0, 365.1, 365.2, 365.5, 365.6, 365.9;	C60: C60.1, C60.2, C60.3, C60.4, C60.5, C60.6, C60.8, C60.9; C61: C61.1, C61.2, C61.3, C61.4, C61.5, C61.8, C61.9; C62: C62.1, C62.2, C62.3, C62.4, C62.8, C62.9;	1277 (“glaucoma”);	1436 (“glaucoma surgery/trabeculectomy”);	None
Hearing loss	H90: H90.0, H90.1, H90.2, H90.3, H90.4, H90.5, H90.6,	389: 389.0, 389.1, 389.2,	NA	NA	NA	None

Outcome	ICD-10	ICD-9	OPCS-4	Non-cancer illness code[1]	Operation code[1]	Exclusion criteria[2]
	H90.7, H90.8; H91: H91.1, H91.3, H91.8, H91.9; Z45: Z45.3; Z46: Z46.1; Z96: Z96.2; Z97: Z97.4;	389.7, 389.9;				
Heart failure	I11: I11.0; I13: I13.0, I13.2; I25: I25.5; I42: I42.0, I42.5, I42.8, I42.9; I50: I50.0, I50.1, I50.9;	425: 425.4; 428: 428.0, 428.1, 428.9;	NA	1076 ("heart failure/pulmonary odema"), 1079 ("cardiomyopathy");	NA	ICD-10 – I42: I42.1, I42.2; ICD-9 – 425: 425.1; Non-cancer illness – 1588 ("hypertrophic cardiomyopathy (hcm / hocm)");
Hemorrhagic stroke	I60: I60.0, I60.1, I60.2, I60.3, I60.4, I60.5, I60.6, I60.7, I60.8, I60.9; I61: I61.0, I61.1, I61.2, I61.3, I61.4, I61.5, I61.6, I61.8, I61.9;	430: 430.9; 431: 431.9;	NA	1086 ("subarachnoid haemorrhage"), 1491 ("brain haemorrhage");	NA	None
Inflammatory bowel disease	K50: K50.0, K50.1, K50.8, K50.9; K51: K51.0, K51.1, K51.2, K51.3, K51.4, K51.5, K51.8, K51.9;	555: 555.0, 555.1, 555.2, 555.9; 556: 556.9;	NA	1461 ("inflammatory bowel disease"), 1462 ("crohns disease"), 1463 ("ulcerative colitis");	NA	None
Ischemic stroke	I63: I63.0, I63.1, I63.2, I63.3, I63.4, I63.5, I63.6, I63.8, I63.9; I64;	434: 434.9; 436: 436.9;	NA	1583 ("ischaemic stroke");	NA	None
Metabolic dysfunction-associated steatotic liver disease	K74: K74.0, K74.2; K75: K75.8; K76: K76.0, K76.9;	571: 571.5, 571.8, 571.9;	NA	NA	NA	ICD-10 – F10: F10.1, F10.2, F10.9; K70: K70.0, K70.1, K70.2, K70.3, K70.4, K70.9; B15: B15.0, B15.9; B16: B16.0, B16.1, B16.2, B16.9; B17: B17.0, B17.1, B17.2, B17.8, B17.9; B18: B18.0, B18.1, B18.9; B19: B19.0, B19.9; K75: K75.2, K75.4; E88: E88.0, E88.1; E78: E78.6, E78.8; E84: E84.0, E84.1, E84.8, E84.9; E83: E83.0; Q44: Q44.7; O26: O26.6; O14: O14.2; G93: G93.7; Z94: Z94.4; ICD-9 – 303: 303.9; 305: 305.0; 571: 571.0, 571.1, 571.2, 571.3; 070: 070.1, 070.3, 070.5, 070.9; 277: 277.62, 277.0; V42: V42.7;

Outcome	ICD-10	ICD-9	OPCS-4	Non-cancer illness code[1]	Operation code[1]	Exclusion criteria[2]
						Non-cancer illness – 1604 (“alcoholic liver disease / alcoholic cirrhosis”), 1408 (“alcohol dependency”), 1496 (“alpha-1 antitrypsin deficiency”), 1155 (“hepatitis”), 1156 (“infective/viral hepatitis”), 1578 (“hepatitis a”), 1579 (“hepatitis b”), 1580 (“hepatitis c”), 1581 (“hepatitis d”), 1582 (“hepatitis e”); Operation – 1454 (“liver transplant”);
Migraine or headache disorder	G43: G43.0, G43.1, G43.2, G43.3, G43.8, G43.9; G44: G44.0, G44.1, G44.2, G44.3, G44.4, G44.8; G50: G50.0, G50.1, G50.8, G50.9;	346: 346.0, 346.1, 346.2, 346.8, 346.9; 350: 350.1, 350.2, 350.8, 350.9;	NA	1265 (“migraine”), 1523 (“trigemminal neuralgia”);	NA	None
Non-AF arrhythmia	I44: I44.0, I44.1, I44.2, I44.3, I44.4, I44.5, I44.6, I44.7; I45: I45.0, I45.1, I45.2, I45.3, I45.4, I45.5, I45.6, I45.8, I45.9; I47: I47.0, I47.1, I47.2, I47.9; I49: I49.0, I49.1, I49.2, I49.3, I49.4, I49.5, I49.8, I49.9;	426: 426.0, 426.1, 426.3, 426.4, 426.5, 426.6, 426.7, 426.9; 427: 427.0, 427.1, 427.2, 427.4, 427.5, 427.6, 427.8, 427.9;	K57: K57.4, K57.6, K57.7; K59: K59.1, K59.2, K59.3, K59.4, K59.5, K59.6, K59.7, K59.8, K59.9; K60: K60.1, K60.2, K60.3, K60.4, K60.5, K60.6, K60.7, K60.8, K60.9; K61: K61.1, K61.2, K61.3, K61.4, K61.5, K61.6, K61.7, K61.8, K61.9;	1487 (“svt / supraventricular tachycardia”), 1486 (“sick sinus syndrome”), 1484 (“wolff parkinson white / wpw syndrome”);	1096 (“pacemaker/defibrillator insertion”), 1548 (“pacemaker insertion”), 1549 (“pacemaker battery change”), 1550 (“defibrillator/icd insertion”), 1551 (“defibrillator/icd battery change”);	None
Osteoporosis	M80: M80.0, M80.00, M80.01, M80.03, M80.05, M80.07, M80.08, M80.09, M80.1, M80.10, M80.2, M80.21, M80.25, M80.28, M80.29, M80.4, M80.40, M80.41, M80.45, M80.46, M80.48, M80.49, M80.5, M80.50, M80.52, M80.53, M80.55, M80.58, M80.8, M80.80, M80.81, M80.82, M80.85, M80.86, M80.87, M80.88, M80.89, M80.9, M80.90, M80.91, M80.92, M80.93, M80.94, M80.95, M80.96, M80.97, M80.98, M80.99; M81: M81.0, M81.00, M81.02, M81.05, M81.08, M81.09, M81.1, M81.10, M81.19, M81.2, M81.20, M81.29, M81.3, M81.30, M81.38, M81.39, M81.4, M81.40, M81.43, M81.44, M81.45, M81.47, M81.48, M81.49, M81.5, M81.50, M81.51, M81.55, M81.58, M81.59, M81.6, M81.65, M81.67, M81.68, M81.69, M81.8, M81.80, M81.85, M81.87, M81.88, M81.89, M81.9, M81.90, M81.91, M81.92, M81.93, M81.94, M81.95, M81.96, M81.97, M81.98, M81.99; M82: M82.0, M82.00, M82.01, M82.06, M82.08, M82.09, M82.1, M82.10, M82.11, M82.13, M82.18, M82.19, M82.8;	733: 733.05, 733.09, 733.13, 733.14, 733.15, 733.16;	NA	1309 (“osteoporosis”);	NA	None
Parkinson's disease	G20; G21: G21.0, G21.1, G21.2, G21.3, G21.4, G21.8, G21.9; G22; G23: G23.0, G23.1, G23.2, G23.3, G23.8, G23.9;	332: 332.0, 332.1;	NA	1262 (“parkinsons disease”);	NA	None
Peripheral artery disease	I70: I70.0, I70.00, I70.01, I70.2, I70.20, I70.21, I70.8, I70.80, I70.9, I70.90; I73: I73.8, I73.9;	440: 440.0, 440.2; 443: 443.8, 443.9;	L21: L21.6; L51: L51.3, L51.6, L51.8; L52: L52.1, L52.2; L54:	1067 (“peripheral vascular disease”), 1087 (“leg	1102 (“fem-pop bypass/leg artery bypass”), 1108 (“leg	None

Outcome	ICD-10	ICD-9	OPCS-4	Non-cancer illness code[1]	Operation code[1]	Exclusion criteria[2]
			L54.1, L54.4, L54.8; L59: L59.1, L59.2, L59.3, L59.4, L59.5, L59.6, L59.7, L59.8; L60: L60.1, L60.2; L63: L63.1, L63.5, L63.9; L66: L66.7; X09: X09.3, X09.4, X09.5;	claudication/ intermittent claudication"), 1088 ("arterial embolism");	artery angioplasty +/- stent"), 1440 ("amputation of leg");	
Schizophrenia	F20: F20.0, F20.1, F20.2, F20.3, F20.4, F20.5, F20.6, F20.8, F20.9; F21; F22: F22.0, F22.8, F22.9; F23: F23.0, F23.1, F23.2, F23.3, F23.8, F23.9; F24; F25: F25.0, F25.1, F25.2, F25.8, F25.9; F28; F29;	295: 295.0, 295.1, 295.3, 295.4, 295.5, 295.6, 295.7, 295.9;	NA	1289 ("schizophrenia");	NA	ICD-10 – F30: F30.0, F30.1, F30.2, F30.8, F30.9; F31: F31.0, F31.1, F31.2, F31.3, F31.4, F31.5, F31.6, F31.7, F31.8, F31.9; F32: F32.0, F32.1, F32.2, F32.3, F32.8, F32.9; F33: F33.0, F33.1, F33.2, F33.3, F33.4, F33.8, F33.9; F41: F41.2; ICD-9 – 300: 300.4; 311: 311.9; 296: 296.0, 296.1, 296.2, 296.3, 296.4, 296.5, 296.6, 296.9;
Thyroid disease	E01: E01.1, E01.8; E02; E03: E03.0, E03.1, E03.2, E03.3, E03.4, E03.8, E03.9; E05: E05.0, E05.1, E05.2, E05.3, E05.4, E05.5, E05.8, E05.9; E06: E06.2, E06.3, E06.5; E89: E89.0;	242: 242.8, 242.9; 244: 244.2, 244.99; 245: 245.2, 245.9;	NA	1225 ("hyperthyroidism/thyrotoxicosis"), 1226 ("hypothyroidism/myxoedema"), 1428 ("thyroiditis"), 1522 ("grave's disease");	NA	None
Type 2 diabetes	E11: E11.0, E11.1, E11.2, E11.3, E11.4, E11.5, E11.6, E11.7, E11.8, E11.9;	250: 250.00, 250.10;	NA	1223 ("type 2 diabetes");	NA	ICD-10 – E10: E10.0, E10.1, E10.2, E10.3, E10.4, E10.5, E10.6, E10.7, E10.8, E10.9; ICD-9 – 250: 250.01, 250.11; Non-cancer illness – 1222 ("type 1 diabetes");
Valvular heart disease	I05: I05.0, I05.1, I05.2, I05.8, I05.9; I06: I06.0, I06.1, I06.2, I06.8, I06.9; I07: I07.0, I07.1, I07.2, I07.8, I07.9; I08: I08.0, I08.1, I08.2, I08.3; I34: I34.0, I34.1, I34.2, I34.8, I34.9; I35: I35.0, I35.1, I35.2, I35.8, I35.9; I36: I36.0, I36.1, I36.2, I36.8, I36.9; I37: I37.0, I37.1, I37.2, I37.8, I37.9; I39: I39.0, I39.1, I39.3, I39.4;	394: 394.0, 394.2, 394.9; 395: 395.1, 395.9; 396: 396.9; 424: 424.0, 424.1, 424.3;	K25: K25.1, K25.2, K25.3, K25.4, K25.5, K25.8, K25.9; K26: K26.1, K26.2, K26.3, K26.4, K26.5, K26.8, K26.9; K27: K27.1, K27.2, K27.3, K27.4, K27.5, K27.6, K27.8, K27.9; K28: K28.1, K28.2, K28.4, K28.5, K28.8, K28.9; K30: K30.1, K30.2, K30.3, K30.4; K34: K34.1, K34.2, K34.3, K34.4, K34.5, K34.6,	1586 ("aortic valve disease"), 1490 ("aortic stenosis"), 1587 ("aortic regurgitation / incompetence"), 1584 ("mitral valve disease"), 1488 ("mitral valve prolapse"), 1489 ("mitral stenosis"), 1585 ("mitral regurgitation / incompetence");	1099 ("aortic valve repair/replacement"), 1100 ("mitral valve repair/replacement"), 1101 ("other valve repair/replacement");	None

Outcome	ICD-10	ICD-9	OPCS-4	Non-cancer illness code[1]	Operation code[1]	Exclusion criteria[2]
			K34.8, K34.9; K35: K35.1, K35.2, K35.3, K35.4, K35.5, K35.6, K35.7, K35.8, K35.9;			
Vascular dementia	F01: F01.0, F01.1, F01.2, F01.3, F01.8, F01.9; I67: I67.3;	NA	NA	NA	NA	None
Venous thromboembolism[5]	I80: I80.1, I80.2; I82: I82.2; I26: I26.0, I26.9;	NA	L79: L79.1; L90: L90.2;	1093 (“pulmonary embolism +/- dvt”), 1094 (“deep venous thrombosis (dvt)”), 1068 (“venous thromboembolic disease”);	NA	ICD-10 – D68: D68.0, D68.1, D68.2, D68.3, D68.4, D68.5, D68.6, D68.8, D68.9; I80: I80.0, I80.3, I80.8, I80.9; I81; I82: I82.0;

Codes as defined in outcome-extraction.R (make_outcome_spec). Deaths: cases were also identified from underlying or contributory causes of death (UK Biobank death registry), matched on the 3-character ICD-10 categories derived from each outcome's ICD-10 list.

[1]UK Biobank self-reported non-cancer illness (data-coding 6) and operation (data-coding 5) codes; the script matches on the text labels shown in quotation marks, the numeric codes were mapped from the UK Biobank codings.

[2]A participant with any record of an exclusion code (any source, any date, including death records) is not counted as a case for that outcome.

[3]No outcome specification in outcome-extraction.R; defined as death from any cause, using the date of death recorded in the UK Biobank mortality registry.

[4]No outcome specification in outcome-extraction.R; defined as death with any ICD-10 “I” code (I00–I99) recorded as the primary (underlying) cause of death.